

M01 - Introduction to MLOps and Autonomous AI Systems



Topic

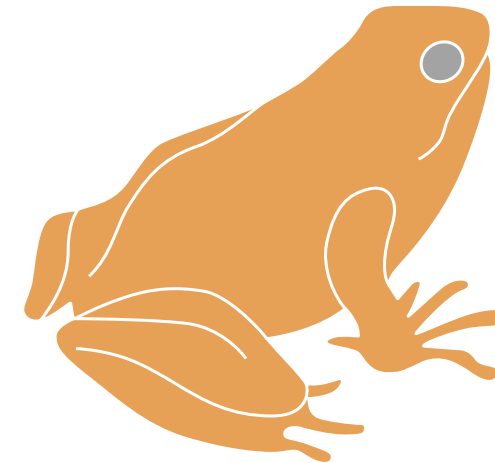
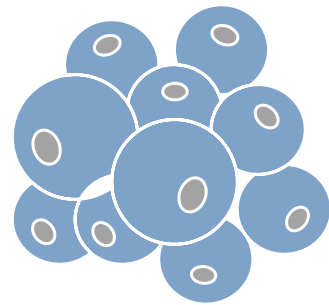
- **Evolution from ML to MLOps**
- **CI/CD/CT for Machine Learning**
- **Challenges of Production AI**
- **AI Lifecycle Management**
- **MLOps Maturity Model**
- **Billing Assurance Agent Overview**
- **Hands-on Discussion**
- **Summary**





Evolution from ML to MLOps

Evolution from ML to MLOps



Traditional ML

Manual model
building and
deployment

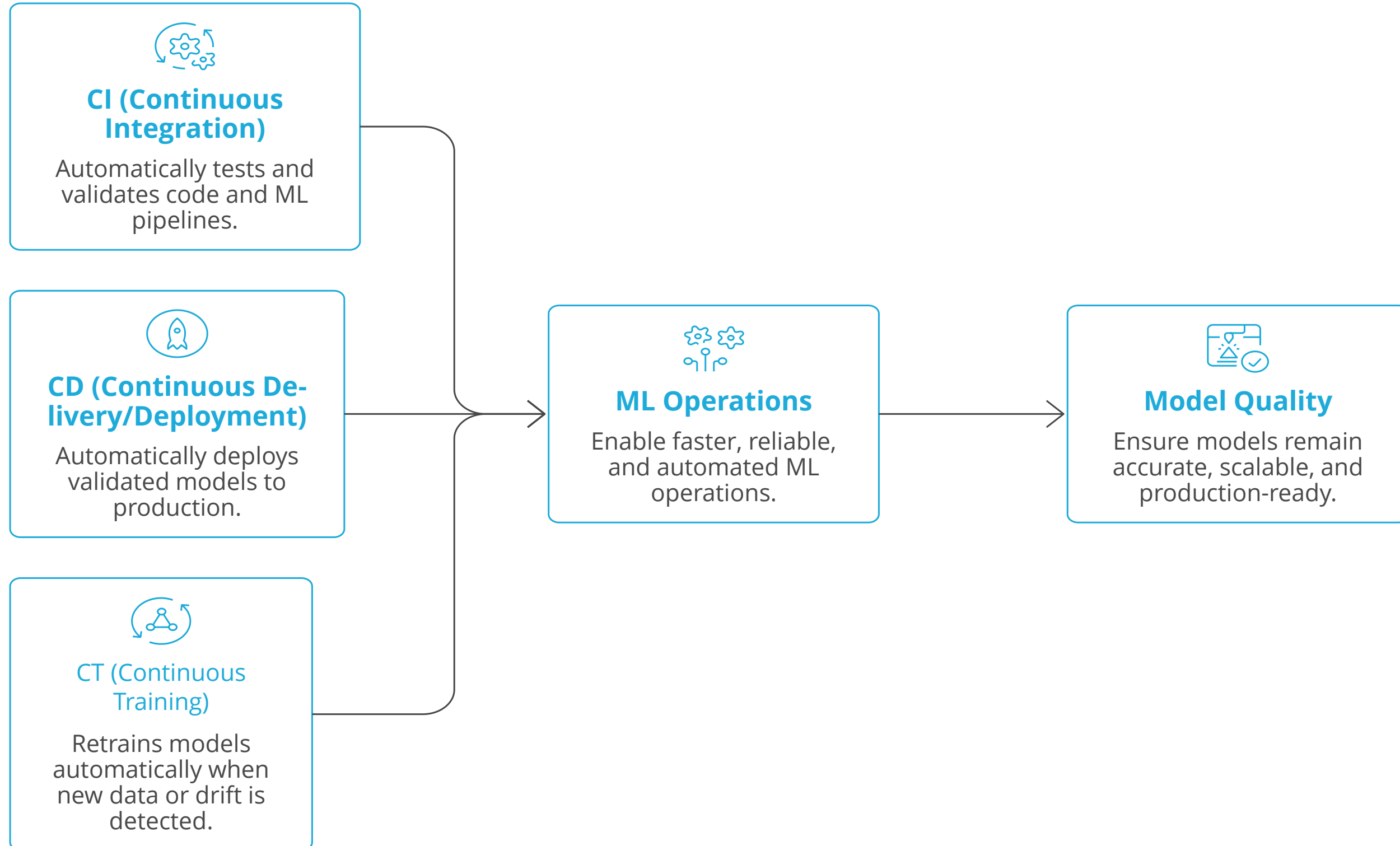
MLOps Adoption

Automating model
lifecycle

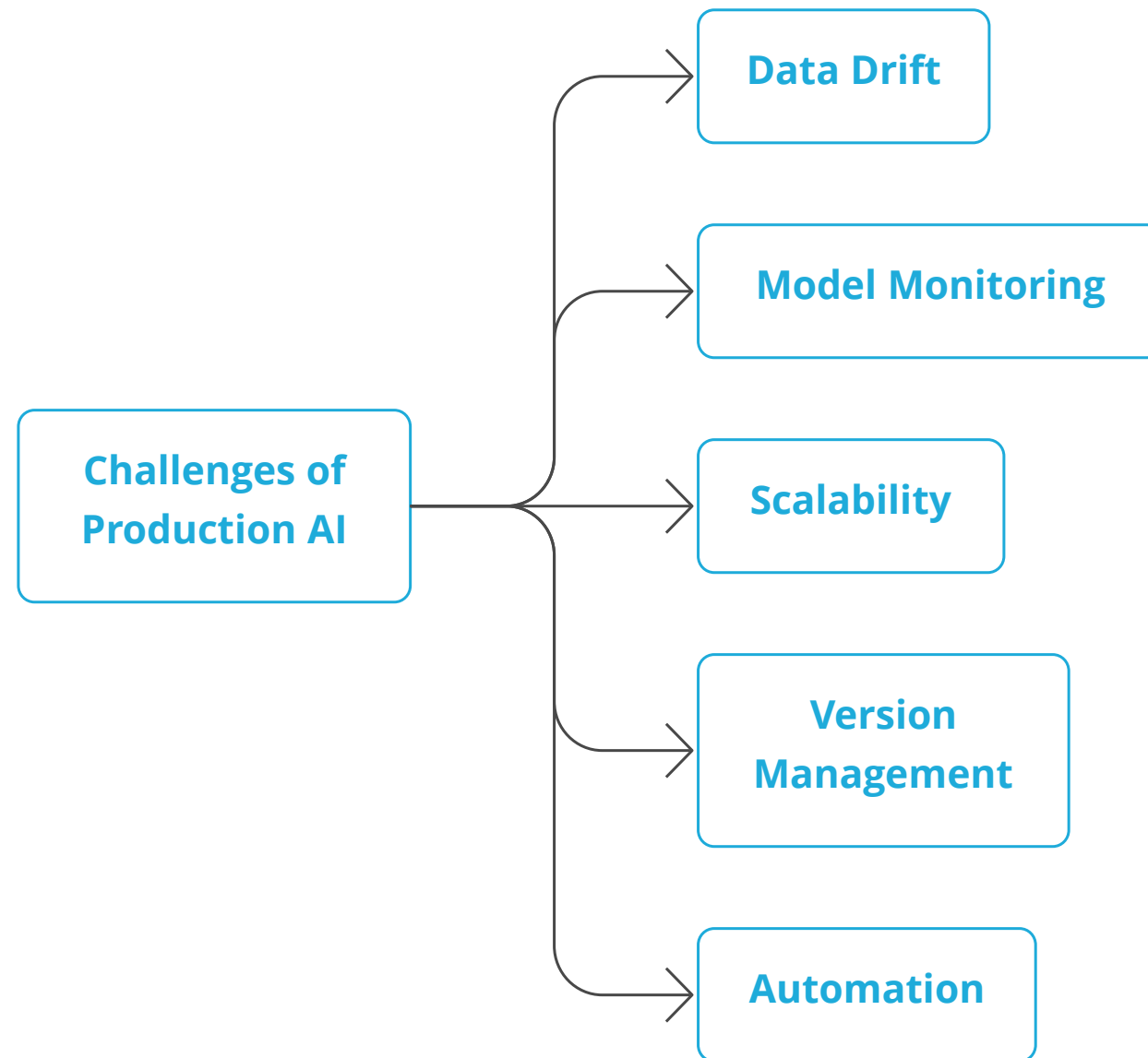
MLOps

Faster, reliable AI
solutions

CI/CD/CT for Machine Learning

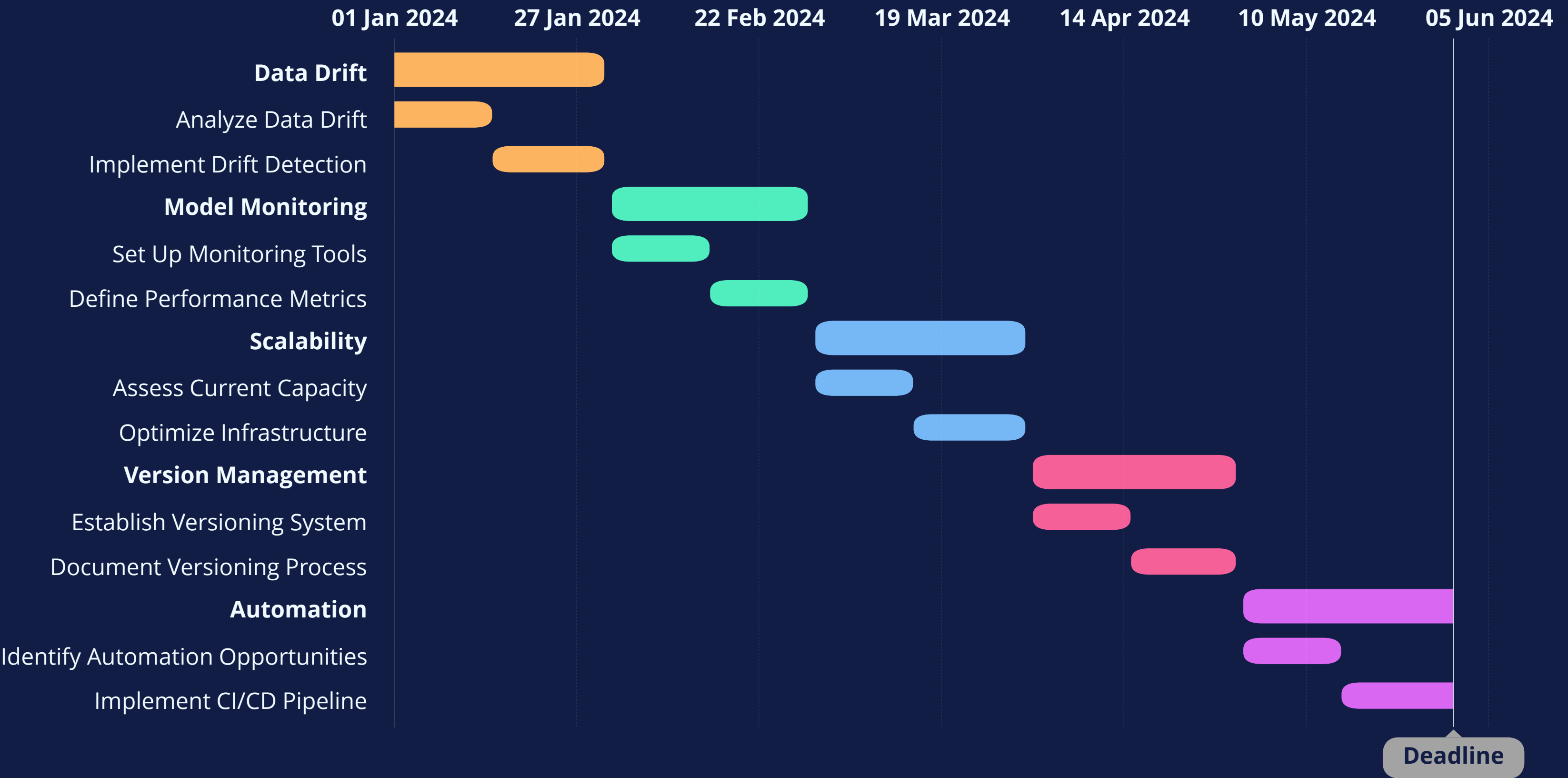


Challenges of Production AI

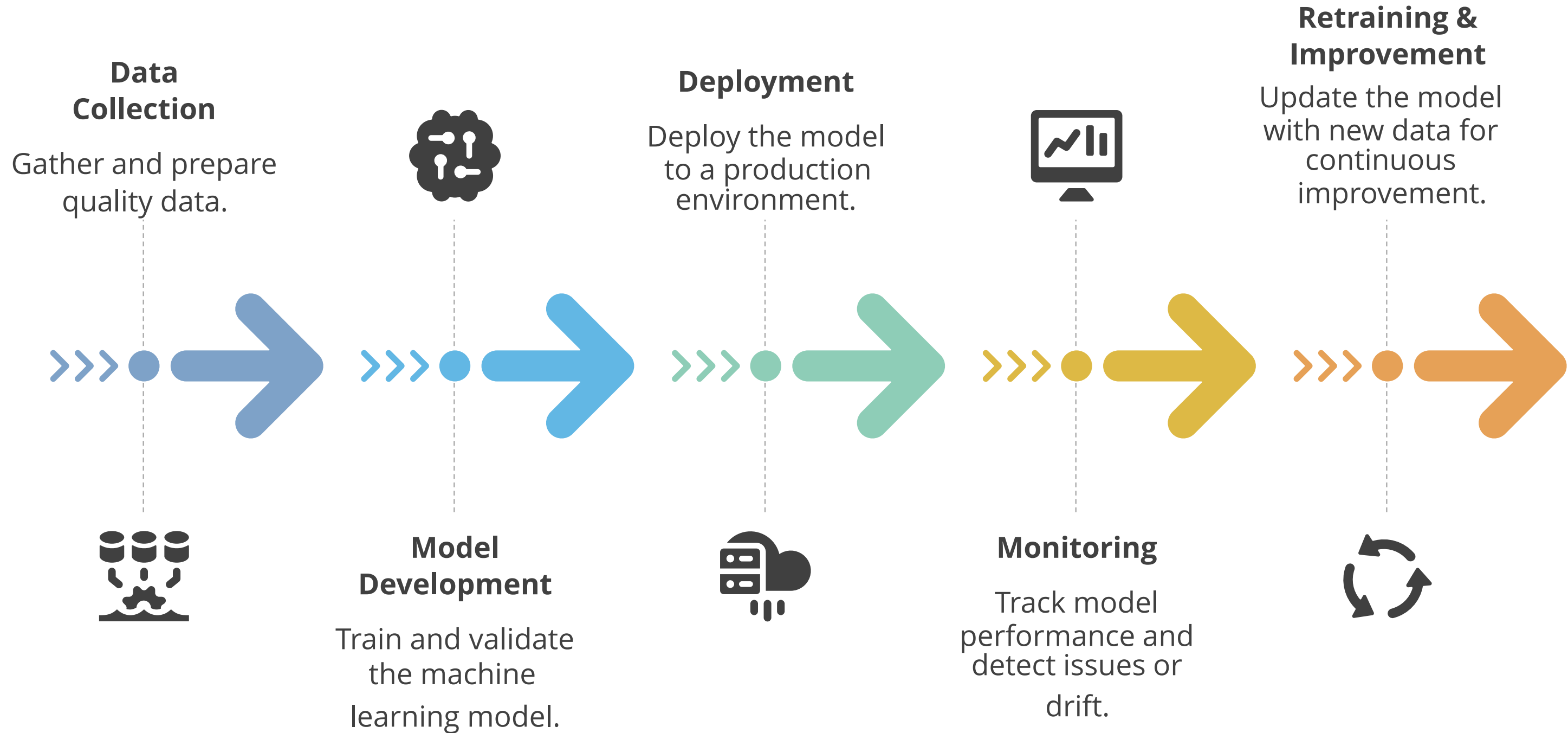


- **Data Drift** – New data differs from training data, reducing model accuracy.
- **Model Monitoring** – Continuous tracking is needed to detect performance issues.
- **Scalability** – Models must handle increasing users and data efficiently.
- **Version Management** – Managing datasets, code, and model versions can be complex.
- **Automation** – Manual training and deployment are slow and error-prone.

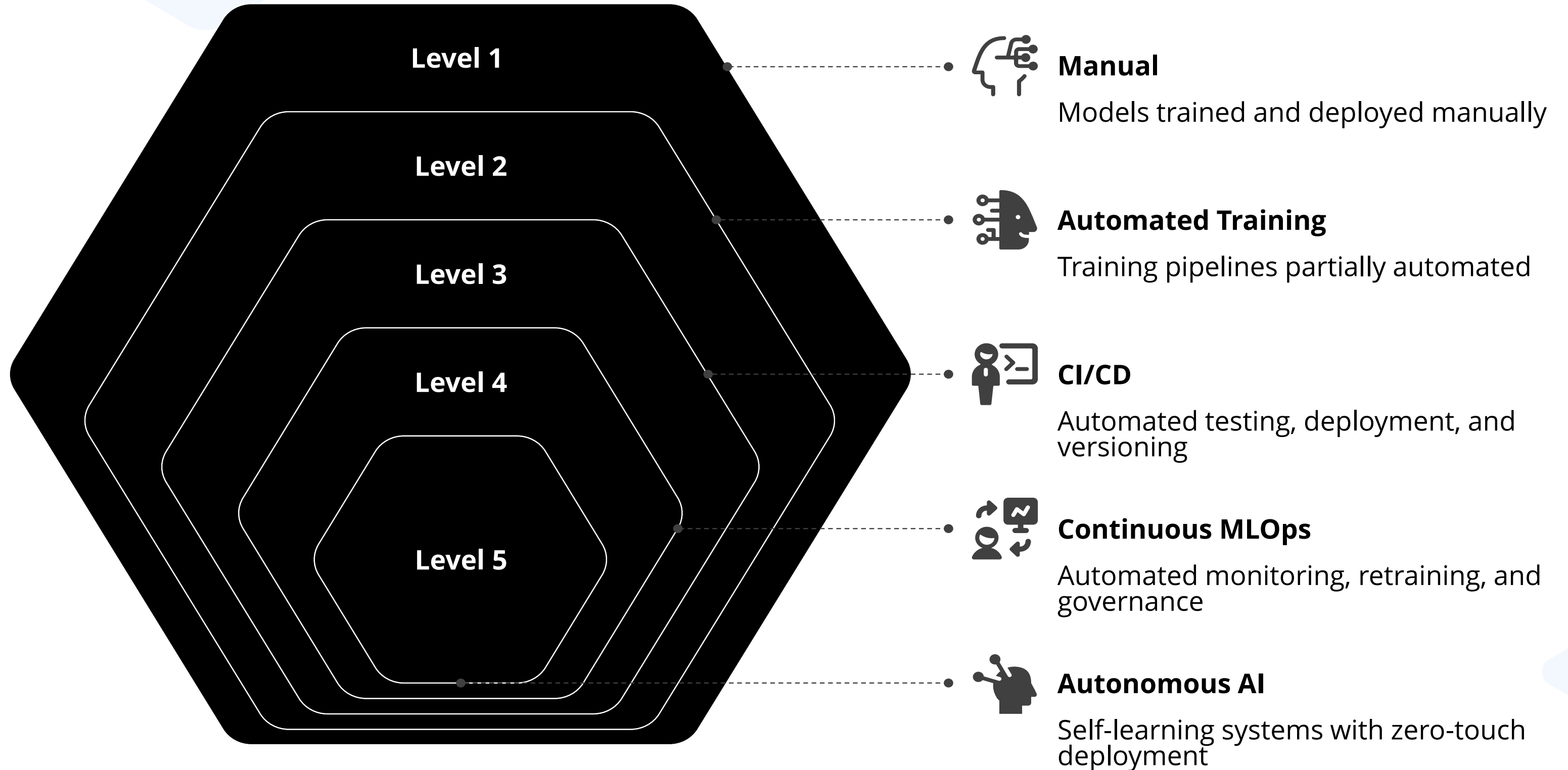
Challenges of Production AI



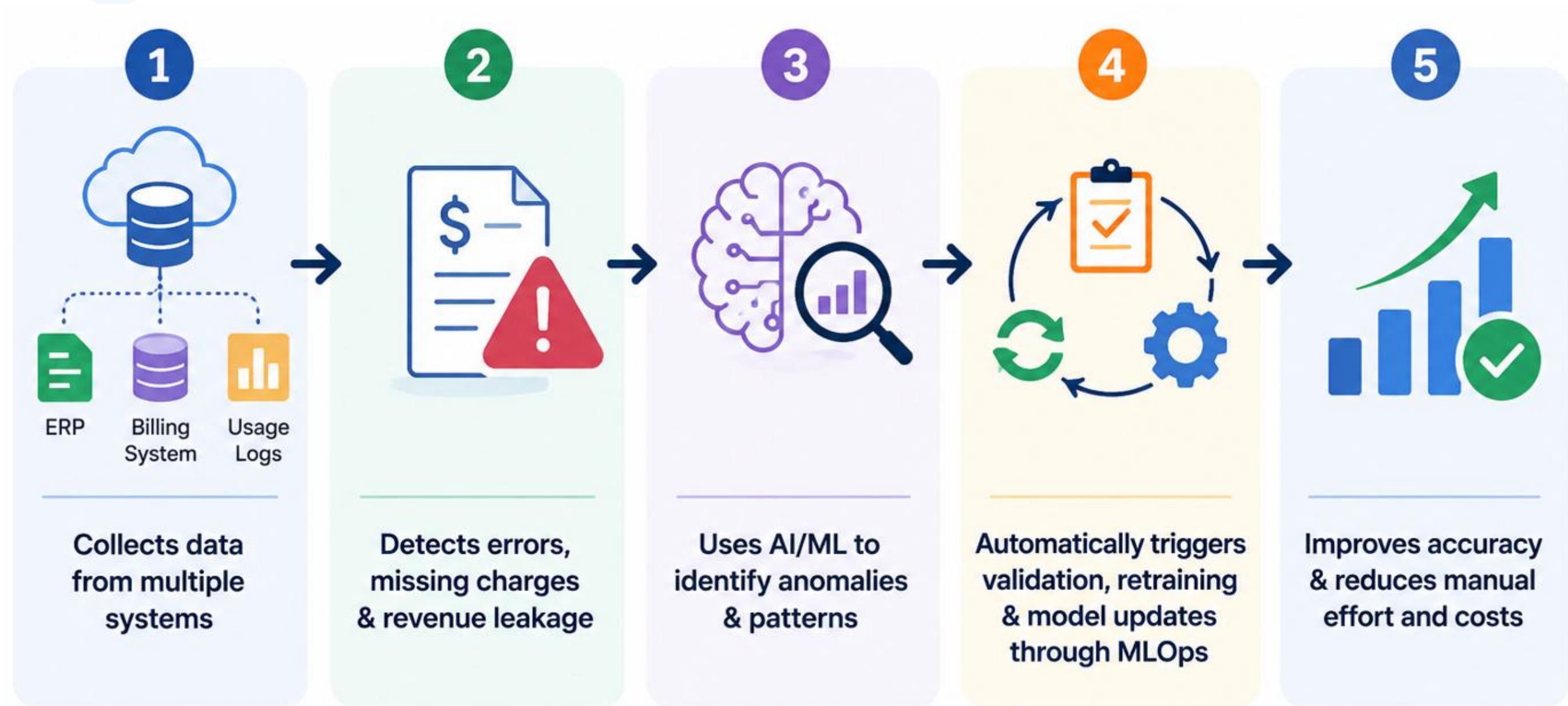
AI Lifecycle Management



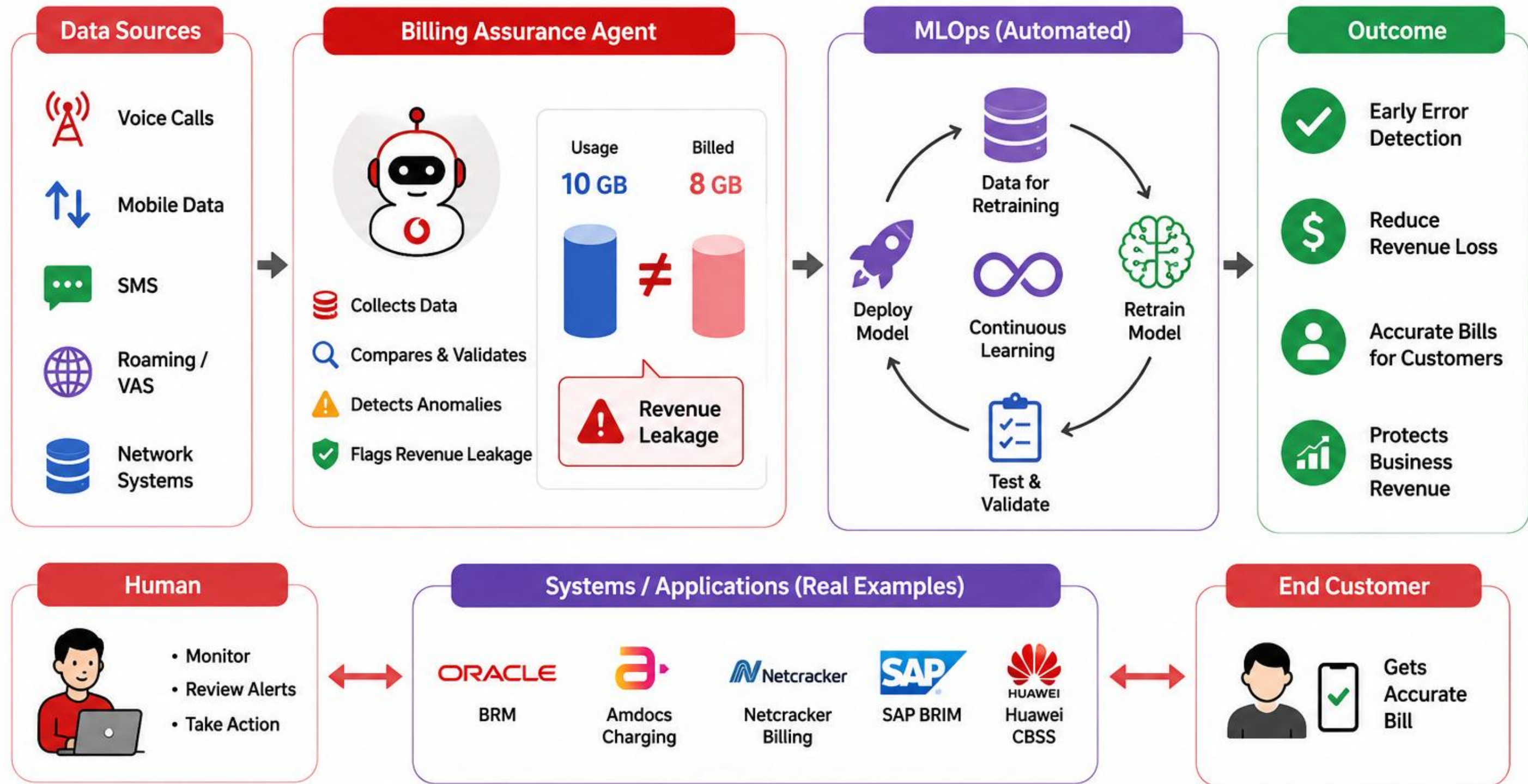
MLOps Maturity Model



Billing Assurance Agent Overview



Vodafone Billing Assurance Agent



Accurate Billing



Operational Efficiency



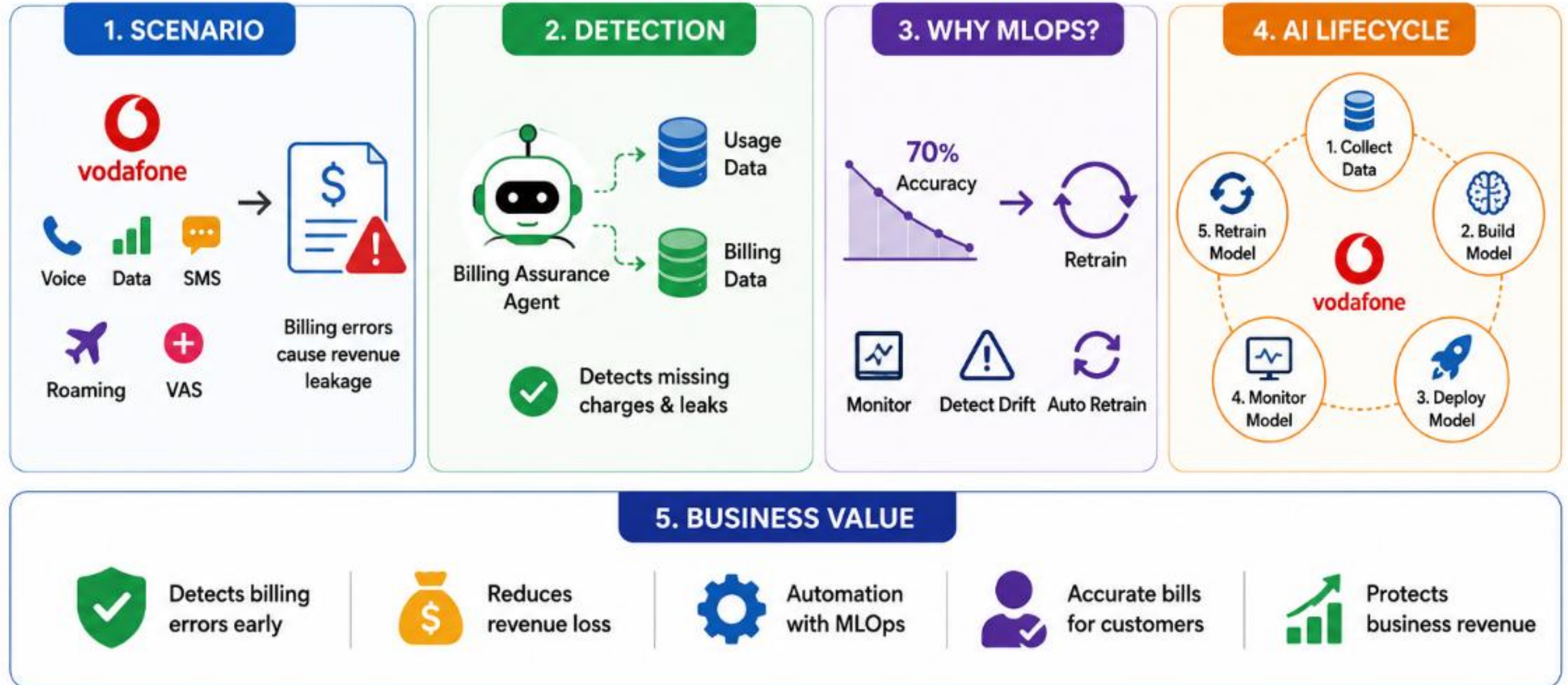
Revenue Protection



Happy Customers

Lab Activity

MLOps Evolution & Billing Assurance Analysis



Student Task: Analyze each scenario using the 5 key questions:

1

Business Problem

2

Is AI Required?

3

Data Needed?

4

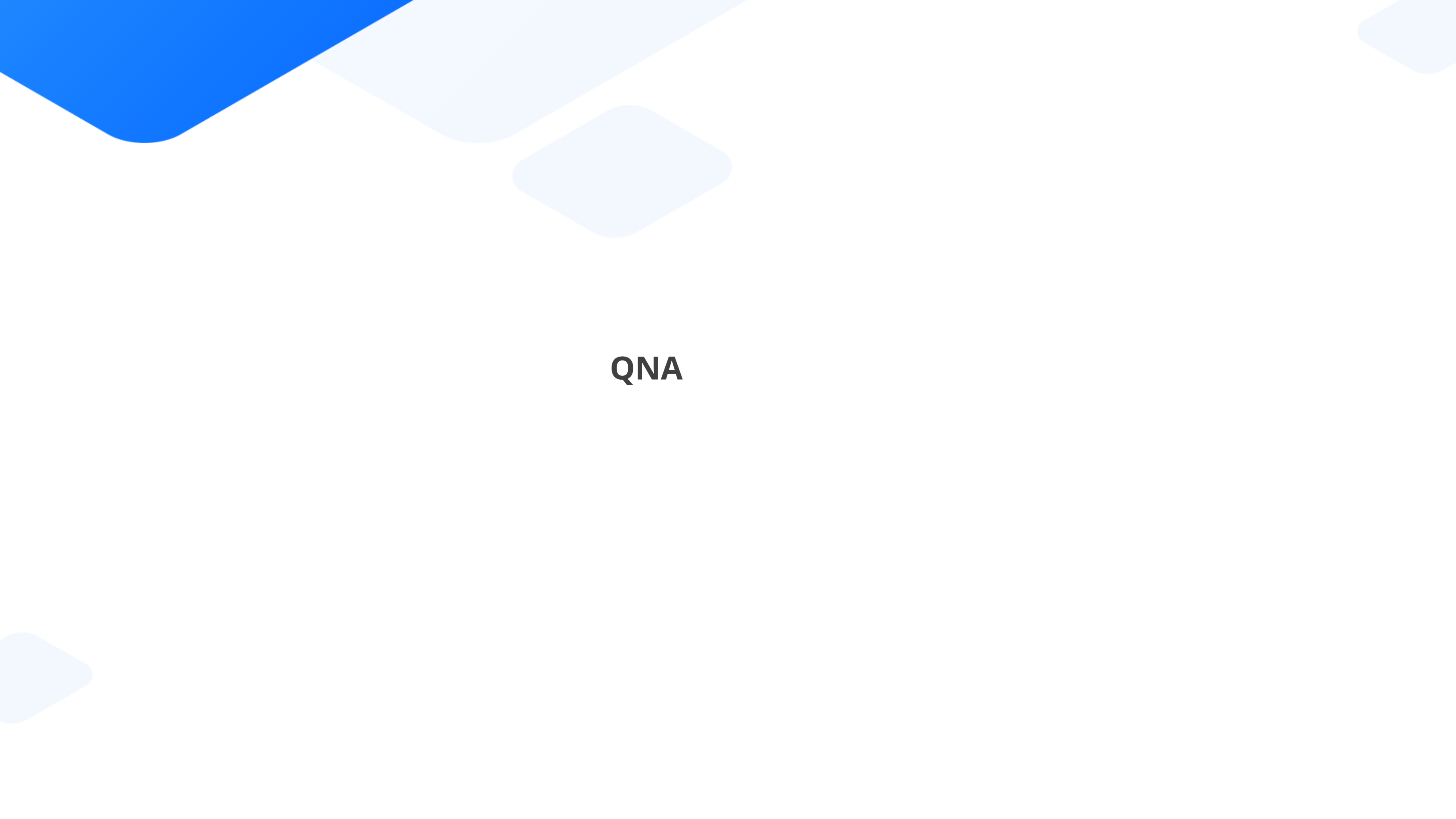
Challenges?

5

How can MLOps help?

Introduction to MLOps and Autonomous AI Systems – Summary

- ✓ Understand the **evolution from Machine Learning (ML) to MLOps**.
- ✓ Learn the role of **CI, CD, and Continuous Training (CT)** in ML.
- ✓ Explore the **challenges of deploying AI models** in production.
- ✓ Understand the complete **AI Lifecycle** from data collection to retraining.
- ✓ Learn the **MLOps Maturity Model**, from manual processes to autonomous AI.
- ✓ Explore a real-world **Vodafone Billing Assurance Agent** use case.
- ✓ Identify how **MLOps improves automation, monitoring, governance, and model reliability**.
- ✓ Discuss real-world business scenarios and analyze **where MLOps adds value**.



QNA